Bagas Adjie Pamungkas

Data Science & AI/ML Engineering Resume

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Profile

Information Engineering undergraduate at Universitas Gadjah Mada building end-to-end ML, deep learning, and agentic LLM systems—from real-time anomaly detection and spatiotemporal forecasting to Hybrid-RAG agents, with strict focus on zero-leakage evaluation and production serving.

Education

Universitas Gadjah Mada | GPA : 3.38 2024 – Expected 2028

Bachelor of Engineering, Information Engineering

Organizational Experience

KMTETI FT UGM – *Staff of Workshop Division* Nov 2024 – Present

Facilitated technical training programs and engineering workshops (Data Science, Web Dev, Microcontrollers).

Find IT! UGM – *Staff of Equipment, Logistics, and Consumption Division* Oct 2024 – May 2025

Coordinated technical procurement, venue readiness, and logistics operations across national IT competition tracks.

Selected Projects

KlaimKilat AI – Omnichannel Reimbursement & Audit Agent GitHub

- Built a stateful LangGraph agent (Web/Telegram/WhatsApp intake) running vision extraction, hybrid RAG SOP matching, and quota/audit tool calls to auto-approve, reject, or escalate claims via human-in-the-loop interrupts on FastAPI + Redis.
- Fused pgvector dense (1024-dim) + Postgres full-text search via RRF ($k = 60$) over table-aware chunks, reaching Recall@4 1.0 and MRR 1.0 across 40 benchmark scenarios, with 0% false auto-approve rate and per-node telemetry.

TransitSentinel – Cashless Transit Anomaly Detection & Audit GitHub

- Engineered a real-time anomaly engine over 120k simulated transit tap transactions (45 stations, 120 readers) using 28 causal spatio-temporal features (Haversine velocity, rolling retry/latency z-scores) with zero temporal leakage.
- Stacked sub-0.1ms deterministic guardrails, PyTorch Residual VAE, and Isolation Forest + LightGBM (TreeSHAP) with cost router ($\tau^* = 0.41$), hitting 0.9868 PR-AUC, 100% refund precision, 98% net savings, and <25ms p95 on FastAPI.

Customer Service Ticket Triage & RAG Copilot GitHub

- Fine-tuned multi-task IndoBERT with PEFT/LoRA (2.41% trainable params) and GroupShuffleSplit leakage checks over 1,200 tickets, lifting Category Macro-F1 to 0.9818, Urgency to 0.9644, and Critical recall from 0.5357 to 1.0000.
- Added OOF temperature scaling ($T = 0.60$) and F_β-tuned threshold for calibrated confidences, deterministic PII masking (F1 1.0), and hybrid BM25 + FAISS Multilingual-E5 SOP retriever (Recall@5 1.0, MRR@5 0.997).

Hierarchical Waste Classification with SigLIP GitHub

- Extracted frozen 1,152-dim SigLIP-SO400M embeddings over a cleaned 28,965-image dataset (70/15/15 split) across 3 primary categories and 8 trained sub-labels, leveraging class weighting, label smoothing, and embedding MixUp.
- Built a cross-family ensemble (Multi-Seed Residual MLP + cosine kNN + RBF SVM), reaching 0.9956 val Macro-F1 (0.9936 test; 0.9958 B3 recall) on primary categories and 0.9591 Macro-F1 on human-verified sub-labels.

Bengawan Solo River Water Level Forecasting GitHub

- Engineered 187 leakage-free physical features across 30 stations (84k train, 888k hourly weather records), combining FAO-56 Penman-Monteith VPD, soil moisture runoff, reservoir lag dynamics, and HydroRIVERS IDW routing.
- Trained a 7-member Huber ensemble (LightGBM, XGBoost, CatBoost, Ridge Fourier) with 4-fold rolling-origin CV and causal decaying-residual anchoring, achieving CV mean RMSE 1.394 (MAE 0.536, -8.2% vs diversified baseline).

Technical Skills

AI/ML, NLP & Computer Vision: Python, PyTorch, Hugging Face, IndoBERT, PEFT/LoRA, Variational Autoencoder (VAE), Isolation Forest, TreeSHAP, SigLIP, Scikit-learn, LightGBM, Optuna, OpenCV

LLM, RAG & Serving: LangGraph, Hybrid RAG (pgvector + FTS/RRF, BM25 + FAISS), FastAPI, Redis, Docker, Git

Data Science & Engineering: SQL, Pandas, NumPy, GeoPandas, Google Earth Engine (GEE), Streamlit

Courses & Training

Training: AWS Academy (Cloud Web Application Builder); Kiro x Agentic AI Workshop (DTETI UGM).

Competitions: Big Data Challenge Satria Data 2026 (0.9962 F1); First Runner Up – Final Project Bootcamp TETI AI (2025); First Runner Up – Final Project Bootcamp TETI MobDev (2026).